

SkillSprint AI - Complete Flow Walkthrough

Sarmad Motors Pakistan (Pvt) Ltd · Automobile Company Scenario · End-to-end operating guide

This document is the guided tour of SkillSprint. It assumes the database is empty except for the seeded administrator (admin@gmail.com / 123456789). It walks step-by-step through the exact order in which the admin creates data, uploads content, generates plans, validates them, and lets employees consume them. Every step explains what the module does, why it matters, and how it feeds the next step in the chain.

Starting state

Only one user exists at startup: the administrator. There are no employees, no job roles, no documents, no requirements, no plans. Any attempt by the admin to generate a plan would fail with "no employee selected" and "no job role available" - the AI has nothing to compose from.

Step 1 - Admin logs in and changes the default password

Admin opens the app, signs in with admin@gmail.com / 123456789, and is prompted (or should be, on first login) to change the default password. This is basic hygiene: the seeded password is publicly documented in AI_USAGE.md.

Purpose: establish trusted admin identity. **Impact:** without an admin, nobody can bootstrap the system.

Step 2 - Admin creates the Job Roles catalogue

Admin navigates to *Job Roles* and creates the fifteen roles listed in 02_JobRoles.pdf: Sales Executive, Service Advisor, Mechanical Technician and so on. For each role, the admin fills in name, department, and a written description of the role's responsibilities.

Why now: plans generate *for a role*, not for a person in the abstract. If roles don't exist first, employees cannot be created because the employee form requires a role selection. **Impact if skipped:** the entire generation pipeline is blocked.

Step 3 - Admin uploads the Document corpus

Admin navigates to *Documents* and uploads the twenty-five PDFs from the Documents/ folder one at a time. For each upload the admin fills doc_code, name, doc_type, department, effective date, and version label. On upload the system parses text, chunks paragraph-by-paragraph, indexes each chunk, and runs the injection detector.

Why now: requirements are anchored to documents, and plans cite documents. Documents are the raw material. **Impact if skipped:** requirements cannot cite anything, and generated plans cannot be validated for traceability - every plan would fail hallucination and traceability checks.

Step 4 - Admin creates the Requirements list

Admin navigates to *Requirements* and creates the twenty entries from 04_Requirements.pdf. Each requirement is bound to a source document (from step 3) and a section. Type, must-type, priority and due stage are set here.

Why now: a requirement without its source document cannot be saved (the source_document_id field is required). **Impact if skipped:** the matrix has nothing to map, and no plan can be generated.

Step 5 - Admin builds the Role Requirement Matrix

Admin navigates to *Role Matrix* and assigns requirements to each role. The baseline (R001, R002, R003, R013, R014) applies to every role; role-specific requirements are added on top as per 05_RoleMatrix.pdf. Each assignment is marked mandatory or optional.

Why now: the AI cannot know which subset of the requirement catalogue applies to a Body Shop Technician versus a Marketing Executive; the matrix answers that. **Impact if skipped:** generation would either fail (no requirements resolvable for the role) or produce an irrelevant plan.

Step 6 - Admin creates employee accounts

Admin navigates to *Users* and creates the twenty employees from 01_Users.pdf. Each employee gets an employee_id, name, email, job role (from step 2), department, experience level and joining date. Default password is password123, which HR asks the employee to change on first login.

Why now: onboarding plans are generated per employee. The employee record needs to exist before generation, and roles must exist first (Step 2 precedence). **Impact if skipped:** generation cannot target anyone.

Alternative flow (if the company opens self-signup): the employee can create their own account and the admin approves it, then assigns a job role. Sarmad Motors uses admin-driven creation for control.

Step 7 - Admin generates the onboarding plans

Admin navigates to *Plans*, clicks *Generate*, picks an employee (say Ahmed Raza Khan, Sales Executive), and confirms the role. The system builds the prompt from the corpus and requirements, sends it to Groq, receives a JSON plan, stores it, and displays it. This takes roughly ten to fifteen seconds for a typical role.

Why now: the plan needs all the preceding data - the AI is composing a plan for a specific employee, using a specific role's requirements, citing specific documents. **Impact if run earlier:** the AI would produce empty or fabricated plans because it has no ground truth.

Fallback: if Groq is unavailable (quota, network), the dispatcher automatically retries on Google Gemini. The generation-run record notes which provider was used.

Step 8 - Admin validates each plan

Admin opens each generated plan and clicks *Validate*. The deterministic Python pipeline runs eight checks: coverage, traceability, hallucination, duplicates, contradiction, role relevance, sequence, distractors. The plan receives one of the ten SRS statuses (verified, verified_with_warning, partially_verified, coverage_gap, ...).

Why now: the plan must be validated before an employee sees it. A hallucinated task or an uncovered mandatory requirement is not something to discover on day one of a new hire's tenure. **Impact if skipped:** plans could be released with fabricated content.

Step 9 - Reviewer resolves findings

The reviewer (in a small dealership, the admin themselves; in a larger organisation, a separate HR reviewer) opens the Reviews page. Each validator finding is either acknowledged (accepted), overridden with justification (recorded false positive), or the plan is regenerated. Every action is audit-logged.

Why now: a machine finding is not the final judgement; a human decides whether the plan is releasable. **Impact if skipped:** a plan with unresolved findings is never released to the employee, so onboarding stalls.

Step 10 - Employee opens the plan and works through it

The employee logs in with their email and the reset password, opens *My Onboarding*, and starts with the first module. They read the linked source sections, complete quizzes (auto-graded), submit assessment rubrics (supervisor-graded), and mark modules complete. Progress is updated in real time.

Why now: the plan exists, is validated, and is reviewed. Everything before this step exists to make this step trustworthy. **Impact:** the employee learns from a plan whose every claim is traceable, verified and reviewed - the Skillsprint contract.

Step 11 - Admin monitors Progress

Admin (or the branch manager) opens *Progress* and sees each employee's overall percent, weak areas, SLA state (on_track / behind_schedule / at_risk) and adaptive recommendations. Employees flagged as at_risk get a follow-up conversation with their supervisor.

Why: onboarding is a 90-day process; progress data drives Monday morning branch reviews. **Impact if skipped:** nobody knows if a new hire is actually learning.

Step 12 - When a policy changes: Policy Impact

Say the HR Policy Manual is updated (new gratuity rule) six months in. Admin uploads a new version against POL-02 in the Documents module. The system diff-compares and lists every plan citing changed sections. Admin selects *Regenerate affected plans*; a fresh plan is issued to each employee and their onboarding continues on the new version.

Why: stale plans erode trust. **Impact if skipped:** employees are onboarded on obsolete rules and the dealership fails its next OEM audit.

Step 13 - Consistency Testing (periodic)

Admin picks any generated plan and clicks *Run consistency test*. The system regenerates the plan and compares run-A to run-B. A composite score below 70 percent triggers a review of the prompt template, temperature or model.

Why: catches AI drift before it becomes a systemic problem. Runs monthly at Sarmad Motors.

Step 14 - Reports (end of month)

Admin exports Coverage, Progress, Validation Findings and Audit reports. Coverage feeds the OEM compliance file; Progress feeds the branch monthly review; Validation Findings feed the internal quality committee; Audit feeds the annual internal audit.

Why: operational data must leave the system as portable evidence.

Step 15 - Cross-Entity Search (ad-hoc)

At any point, admin can go to *Search* and query across users, roles, documents, requirements, plans and the audit log with a single keyword. Useful for questions like *which plans cite SOP-04, which employees have not completed R013, which reviewer overrode findings on hallucination in the last quarter*.

Why: operational recall - without it, the tool is a black box.

Recap of the strict order

#	Step	What it unlocks
1	Admin logs in and changes default password	Trust the admin identity
2	Create Job Roles (15)	Employees cannot exist without a role
3	Upload Documents (25)	Requirements cite documents
4	Create Requirements (20)	Matrix rows point at requirements
5	Build the Role Matrix (~70 assignments)	Bridges roles and requirements
6	Create Employees (20)	Plans generate for employees
7	Generate Plans	Needs everything above
8	Validate each Plan	Deterministic checks before release
9	Review findings	Human sign-off
10	Employee consumes the plan	The product in use
11	Track Progress	Onboarding is a journey, not an event
12	Policy Impact when documents change	Keep plans current
13	Consistency Testing (monthly)	Catch AI drift
14	Reports (monthly / on demand)	Portable evidence
15	Search (ad-hoc)	Operational recall

What each page in the app is for

Page	Purpose
Login	Authenticates the user. Uses HttpOnly JWT cookies with SameSite=None for cross-site production
Admin → Dashboard	One-glance status: plans generated, plans awaiting review, employees at risk, documents needing
Admin → Users	CRUD employees. Also lets admin trigger a password reset link.
Admin → Job Roles	CRUD roles. Once a role has employees or matrix rows, it cannot be deleted, only inactivated.
Admin → Documents	Upload, version, list, view chunks, view adversarial flags. Superseding a doc kicks off Policy Impact
Admin → Requirements	CRUD requirements. Each must cite a source document.
Admin → Role Matrix	Add or remove requirements per role. Marks mandatory versus optional.
Admin → Plans	List all plans; filter by employee, role, status. Generate new. Regenerate. View history.
Admin → Plan Detail	Read the plan as the employee sees it. Buttons: Validate, Consistency Test, Reviews, Regenerate
Admin → Reviews	Queue of open review items across all plans. Take action - acknowledge / override / regenerate / c
Admin → Consistency	Run a two-run comparison on any plan; view historical scores.
Admin → Impact	Superseded documents and the plans they touched; regenerate in bulk or individually.
Admin → Search	Cross-entity search bar plus filters.
Admin → Reports	Pick a report kind, date range and format (CSV / PDF); download.
Admin → Audit	Read-only, searchable, filterable audit log.
Employee → My Onboarding	The employee's own plan. Modules, tasks, quizzes, assessment rubric.
Employee → Progress	The employee's own progress percent and weak areas.
Employee → Profile	Password change, profile edit within permitted fields.

Fail-safe reminders

- Never skip a step - the system tolerates missing optional fields but a role, requirement or document missing at a lower step will block everything downstream.
- Every mutation is audit-logged. Do not treat the audit trail as optional - the OEM auditor and the internal audit committee both consume it.
- Password rotations, MFA setup and lost-device reporting apply to SkillSprint the same way they apply to any other business system at Sarmad Motors.
- The *data/* folder next to this walkthrough carries the seed content. Use it to bootstrap; edit it in the app after the first login.